

Question	Marking Guidance	Mark	Comments
01.1	1. The (individual) chromosomes are visible because they have condensed; 2. (Each) chromosome is made up of two chromatids because <u>DNA</u> has replicated; 3. The chromosomes are not arranged in homologous pairs, which they would be if it was meiosis;	2 max	Both parts of each answer are required – evidence and explanation. 1. For 'they' accept 'chromosomes/chromatin/DNA' 1. Accept 'tightly coiled' or 'short and thick' for condensed but do not accept 'contracted'. 2. Accept 'sister chromatids' for 'two chromatids'. 3. Accept not meiosis because bivalents/chiasmata/crossing over not seen. Ignore references to nucleus/nucleolus/nuclear membrane.
01.2	Automarked q – ☒ prophase	1	
01.3	1. Water moves into the cells/cytoplasm by <u>osmosis</u>; 2. Cell/cytoplasm gets bigger;	2	1. Reject water moving into chromosomes/nucleus. 2. Accept idea of cell/cytoplasm has greater volume/swells/expands. 2. Ignore references to pressure changes, turgidity and chromosomes being more dilute. 2. Ignore references to changing water/fluid contents of the cell. 2. Allow ECF for 'nucleus expands' but not for 'chromosomes expand'.

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01.4	Differences in base sequences OR Differences in histones/interaction with histones OR Differences in condensation/(super)coiling;	1	Answer must be in context of differences in arrangement of chromosomes not just related to the properties of the stain. Accept spec section 8 ideas e.g. different methylation/acetylation Accept different genes Reject different alleles
01.5	(Two chromosomes that) carry the same genes;	1	Reject 'same alleles' Accept 'same loci' (plural) or 'genes for the same characteristics'
01.6	(Prokaryotic DNA) is 1. Circular (as opposed to linear); 2. Not associated with proteins/histones ; 3. Only one molecule/piece of DNA OR present as plasmids;	2 max	Max 1 if prokaryotic DNA only found as plasmids OR if prokaryotic DNA is single stranded. Ignore references to nucleus, exons, introns or length of DNA. Do not credit converse statements. Ignore descriptions of eukaryotic DNA alone.

